A Hyperspectral Image NSST-HMF Model and Its Application in HS-Pansharpening

Xianghai Wang, Zhenhua Mu, Ruoxi Song, Jingzhe Tao, and Chuanming Song

***Abstract*— The high spectral resolution of hyperspectral (HS) images provides the possibility of omnidirectional feature identification of objects. However, the high-dimensional features and the high redundancy information properties make data processing and the application of HS images extremely challenging. Thus, effectively expressing and correlating the intrinsic correlations of HS images by establishing a statistical model is of great significance. This article proposes a nonsubsampled shearlet transform hidden Markov forest (NSST-HMF) model. This new approach has three key characteristics: 1) the statistical properties of the NSST coefficients are studied in the spatial and spectral directions, respectively, and the "clustering" and "aggregation" properties are observed in both directions; 2) the HMF structure is proposed to depict the multidimensional collaborative correlation of the HS image NSST coefficients, and the proposed method considers the multidirectional transfer relationships among the Markov structure of HS images NSST coefficient for the first time, which significantly improves the prediction ability of the model; and 3) a novel HS-pansharpening approach based on the NSST-HMF model and amplitude modulation of large state probability in the high-frequency subband direction region is proposed. Experimental results show that our method can efficiently improve the spatial resolution of HS images while simultaneously preserving their spectral features. The HMF structure is first proposed in this article, which provides a way to depict the collaborative correlation of multichannel images.**

***Index Terms*— Hidden Markov forest (HMF) model, hyperspectral (HS) image, HS-pansharpening, nonsubsampled shearlet transform (NSST), spatial–spectral collaborate correlation.**

I. Introduction

As PART of the great revolution of satellite technology that observes Earth, hyperspectral (HS) images can depict the fine spectral features of objects by the solar electromagnetic waves reflected through airborne or spaceborne spectrometers, which lays a good foundation for the quantitative analysis of feature information [1], [2]. However, due to the intrinsic limitations of the sensors, it is difficult to obtain remote sensing images with high spectral resolution and high spatial resolution at the same time. In this way, although the high spectral resolution property of HS images can enhance the discrimination ability of object features, its limitation in spatial resolution greatly affects the accuracy of geometric positioning of the objects [3]. There are several ways to enhance the spatial resolution of HS images, such as HS super-resolution [4], [5], HS–multispectral (MS) fusion, and HS-pansharpening. With the rapid development of HS technology, the newly developed remote sensing platforms can simultaneously acquire the HS images, panchromatic (PAN) images, and MS images. For example, the Earth observation satellite-1 (EO-1) is equipped with the HS imaging spectrometer Hyperion and Advanced Land Imager (ALI), which can obtain the 30-m SSI HS image (220 bands, 0.4–2.5 nm), the 30-m SSI MS image, and the 10-m PAN image at the same time. Thus, the use of software methods to integrate the HS images with high spatial resolution remote sensing of images to achieve information complementation has attracted a great deal of attention from the research community and has become an important research field in HS image processing [6], [7].

As an important branch of multisource remote sensing image fusion [8], the aim of the fusion of HS images with high spatial resolution images is to improve the spatial resolution of the HS images, thus enhancing the visual perception capability of the fused image, which is of great theoretical and practical values for the subsequent applications, such as change detection, feature enhancement, and feature classification of HS images. In the past decades, researchers have studied the methods of fusing MS images with PAN images (referred to as MS-pansharpening) [9], [10], where the high precision orientation properties of the PAN image are used to improve the spatial resolution of the MS image. This improvement has met the application requirements of MS images with high spatial resolution to some extent, such as mapping, stereo-photogrammetry, geology, and flood monitoring. With the continuous development of HS remote sensing platforms, more and more HS-based pansharpening techniques (referred to as HS-pansharpening) [3], [11], [12] have been proposed, including component substitution (CS) [13]–[15], multiresolution analysis (MRA) [16], [17], hybrid method [18], Bayesian method [19], [20], and matrix factorization-based approaches [21], [22]. However, these methods are mostly developed on the basis of MS-pansharpening techniques. Compared with MS-pansharpening, the key challenge of HS-pansharpening is to keep the spectral features of the HS image unchanged. On the one hand, a HS image has

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approximately continuous spectral resolution. To keep this property of HS images after fusion, it is difficult to obtain satisfactory results by treating each band as independent individuals, as in MS-pansharpening. Thus, effectively exploiting and utilizing the correlation of HS bands in HS-pansharpening is essential to improve the pansharpening result. On the other hand, unlike MS and PAN images, which have almost the same spectral range, the spectral range of HS images spans from visible light to shortwave infrared (SWIR) spectra. Because PAN images are only within the visible light spectrum, the large spectral mismatch range brings great difficulty in maintaining the spectral information in the HS-pansharpening process. At present, HS-pansharpening remains a challenging problem [6], [11].

The wavelet transform is a successful tool to capture singular points of the images; inspired by wavelet, more and more multiscale geometric analysis (MGA) methods have been developed to represent the higher dimensional singular features of the images [23], [24]. Their multidirectional and multiscale properties provide the possibility of achieving the sparse representation of images. Among them, the nonsubsampled multiscale transforms, such as the nonsubsampled wavelet transform (NSWT) [25], the nonsubsampled contourlet transform (NSCT) [26], and the nonsubsampled shearlet transform (NSST) [27] have shown significant advantages in many image application fields such as image enhancement, image fusion, and image watermarking [28]–[30], due to their useful shift-invariance property. The HS-pansharpening methods based on MGA have attracted much attention from researchers. Generally, the multiscale transforms are used to decompose the PAN image to extract the detailed information, and then inject them into HS images to improve the spatial resolution. These kinds of methods have the advantages of temporal coherence and image aliasing robustness [11]. However, the methods still lack the ability to keep the spectral information of HS images.

At the same time, the success of MGA in the field of image processing largely comes from the utilization of the transfer and correlation relationship among the subbands, which can not only reveal the intrinsic properties of the image but also provide an effective constraint for maintaining the essential properties of the images. In recent years, research on hidden Markov models (HMMs) in the nonsubsampled multiscale transform domains shows an active trend. Examples include the nondecimated wavelet hidden Markov chain (NWHMC) based on NSWT [31] and the hidden Markov tree model with associated multistate coefficients based on NSST (M-NSST-HMT) [32]. The above-mentioned methods can achieve relatively promising results, but there are still many problems that need to be further studied, such as effective modeling of the multiscale transform coefficients, prediction paths between coefficients, and application fields of models. The HS image containing both kinds of spectral information reflects the target radiation properties and the image geometric information. This kind of image reflects the geometric relationship of a target object, which has a high-dimensional feature and a high redundancy information property. Thus, the method effectively expresses and correlates the intrinsic correlations between the HS image MGA coefficients, while at the same time establishing the corresponding statistical model and is of great significance for improving the quality of HS image information processing. Current methods can only depict the transfer relationship of images with two dimensions but cannot depict the multidirectional transfer relationship of images with higher dimension well, such as video sequence and HS images.

To deal with the above issues in this article, an HS image pansharpening method based on the HS image NSST hidden Markov Forest (HMF) model is proposed. First, the statistical properties of HS image NSST coefficients and PAN image NSST coefficients are studied, including the marginal statistical properties and joint statistical properties in the spatial direction and the correlation between the subbands with the same direction in the spectral direction. We observe that the coefficients in each subband exhibit a sharp peak at zero amplitude and two heavy tails on both sides of the peak. In addition, the "clustering" property and "persistence" property are observed in the spatial and spectral directions. On this basis, the correlation between the hidden states of the HS image NSST subbands in the spatial dimension and the spectral dimension are combined to propose the NSST-HMF for HS images. The combination of the multidimensional correlation of HS images greatly improves the prediction accuracy of the model. At the same time, a parameter estimation method for the model is proposed in combination with the fast classification theory, which improves the computational efficiency of the model to some extent. Furthermore, a HS-pansharpening method is proposed based on the proposed NSST-HMF model, which can overcome the mismatch between the HS and PAN spectral ranges by adaptively modulating the PAN image based on HS bands. The proposed method is compared with some classical pansharpening methods; experimental results demonstrate the effectiveness of the proposed method.

The main contributions of this article are listed as follows.

1) The HS NSST-HMF model which depicts the multidimensional intrinsic correlations of HS images is proposed for the first time.
2) A fast parameter estimation method is proposed to estimate the parameters of the NSST-HMF model.
3) An NSST pansharpening method is proposed based on the NSST-HMF model and direction template.

The rest of this article is organized as follows. Section II gives statistical analysis of NSST coefficients. Sections III and IV are the proposed NSST-HMF model and pansharpening method, respectively. Experimental results and analyses on different data sets are provided in Section V, and Section VI gives the conclusion and future works of this article.

II. Statistical Analysis of NSST Coefficients of HS Images

A. Nonsubsampled Shearlet Transform

NSST is a kind of nonsubsampled multiscale transform [27], [33] which was proposed based on the basis of the shearlet transform [34]. NSST decomposes an image into multiple scales with multiple directions by multiscale and multidirectional decompositions. First, the nonsubsampled

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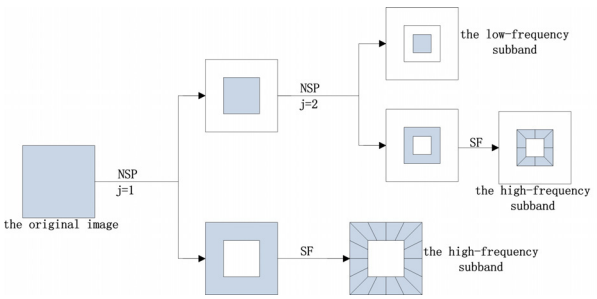


Fig. 1. NSST decomposition process.

pyramid (NSP) is used as the multiscale decomposition filter to decompose the original image into a low-frequency subband and a high-frequency subband. If the decomposition is performed again, the low-frequency subband obtained in the previous stage is decomposed. Therefore, j scale decomposition of an image can produce $j+1$ subbands of the same size as the original image. Then, the high-frequency subbands are decomposed by a modified shearing filter (SF) [35] to obtain the multidirectional subbands. Since there is no subsampled process for the NSP and SF, the NSST is shift-invariant. Fig. 1 shows the decomposition process of NSST.

B. Marginal Statistical Property of NSST Coefficients

To analyze the marginal statistical property of the HS image NSST coefficients, the band images of the HS image Cuprite taken by the airborne visible infrared image spectrometer (AVIRIS) satellite are decomposed by a three-scale NSST. The decomposition directions of each scale are 2, 4, and 8 from coarser to finer. Then, the gray level histograms of the finest subbands are studied. Fig. 2 shows the statistical results for the 3rd band, the 9th band, and the 71st band of the HS image Cuprite. It can be observed in Fig. 2(c) that the probability distributions of the HS band image NSST coefficients exhibit a sharp peak at the zero amplitude and two heavy tails on both sides of the peak, i.e., the number of large-valued coefficients of the image edge region is relatively small, which exhibits the "peak" property, while the number of small-valued coefficients representing the smooth region of the image is relatively large, which exhibits the "tail" property. Similar results can be found in other high-frequency subbands and other test images [32].

Based on the above properties, the Gaussian mixture model (GMM) is used to model the high-frequency NSST coefficients. The "large" state of the GMM represents the large-value coefficient, and the "small" state represents the coefficients with small amplitude. In this way, the NSST coefficients can be depicted by a two-state GMM. The specific modeling process is as follows:

$$\begin{cases} f_{N_i}(Ni) = \sum_{m=1}^{n} f_{S_i}(m) f_{N_i|S_i}(Ni/S_i = m) \\ f_{N_i|S_i}(N_i|S_i = m) = \frac{1}{\sqrt{2\pi\sigma_{im}^2}} \exp\left[-\frac{(N_i-\mu_{im})^2}{2\sigma_{im}^2}\right] \\ \qquad\qquad = g\left(N_i, \mu_{im}, \sigma_{im}^2\right) \end{cases} \tag{1}$$

where $X = \{x_1, x_2, \ldots, x_N\}$ denotes the high-frequency NSST coefficients, S_i is the hidden states, and $f_{S_i}(m)$ is the root probability of each state, which satisfies $\sum_{m=1}^{n} f_{S_i}(m) = 1$, and $f_{N_i|S_i}(Ni/S_i = m)$ represents the corresponding Gaussian probability density distribution function of NSCT coefficients in different states. μ_{im} and σ_{im}^2 represent the mean and variance of the Gaussian distributions, respectively.

Fig. 2(d) shows the fitting results based on the GMM distribution, where the blue line represents the NSST high-frequency coefficient curve, the green line represents the "large" and "small" states of Gaussian distribution curves, and the red line represents the resulting curve fit by the GMM. It can be seen that the GMM can achieve satisfactory fitting performance of the NSST coefficients.

C. Joint Statistical Analysis of NSST Coefficients in the Spatial Direction

The generalized neighborhood relationship of the NSST coefficient X in the spatial direction is shown in Fig. 3 [32]. The coefficient located in the same direction but in the immediate coarser scale of X is defined as the parent coefficient (PX), the eight adjacent coefficients around X that are located in the same direction and the same scale with X are defined as neighborhood coefficients (NX), and the coefficients that are with the same location and scale but at different direction of the reference coefficient X are defined as cousin coefficients (CX).

To analyze the joint statistical properties of NSST coefficients, correlation evaluation methods such as scatter plot, chi-plot, multivariate frequency histogram, and mutual information are used to depict the generalized correlation of NSST coefficients. Fig. 4 shows the statistical results of the eighth direction subband of the finest scale of the third band of Cuprite.

1) Scatter Plot: Fig. 4 shows the scatter plot results. It can be seen from the statistical results that not only the correlation between X and NX shows a strong "clustering" property but that the property also exists between X and PX.

2) Chi-Plot: The chi-plot method is used as a graph method for measuring the correlation of variables. The specific definition of a chi-plot is defined as [36].

Let $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ be the random variable sample, for each data point $(x_i, y_i)(i \in \{1, 2, \ldots, n\})$

$$\begin{cases} Hi = \dfrac{\sum_{j\neq i} I(xj \le xi, yj \le yi)}{n-1}, \quad Fi = \dfrac{\sum_{j\neq i} I(xj \le xi)}{n-1} \\ Gi = \dfrac{\sum_{j\neq i} I(yj \le yi)}{n-1}, \quad Si = \operatorname{sign}\left\{\left(Fi - \dfrac{1}{2}\right)\left(Gi - \dfrac{1}{2}\right)\right\}. \end{cases} \tag{2}$$

According to (2), we can get

$$\begin{cases} \chi_i = \dfrac{Hi - FiGi}{\sqrt{Fi(1-Fi)Gi(1-Gi)}} \\ \lambda_i = 4Si \max\left\{\sqrt{Fi - \dfrac{1}{2}}, \sqrt{Gi - \dfrac{1}{2}}\right\} \end{cases} \tag{3}$$

where $I(A)$ is an indicator function, χ_i is the nonparametric metric for the correlation of sample data, λ_i is the distance from (x_i, y_i) to the sample data center, and the values of χ_i and λ_i are both in the range $[-1, 1]$.

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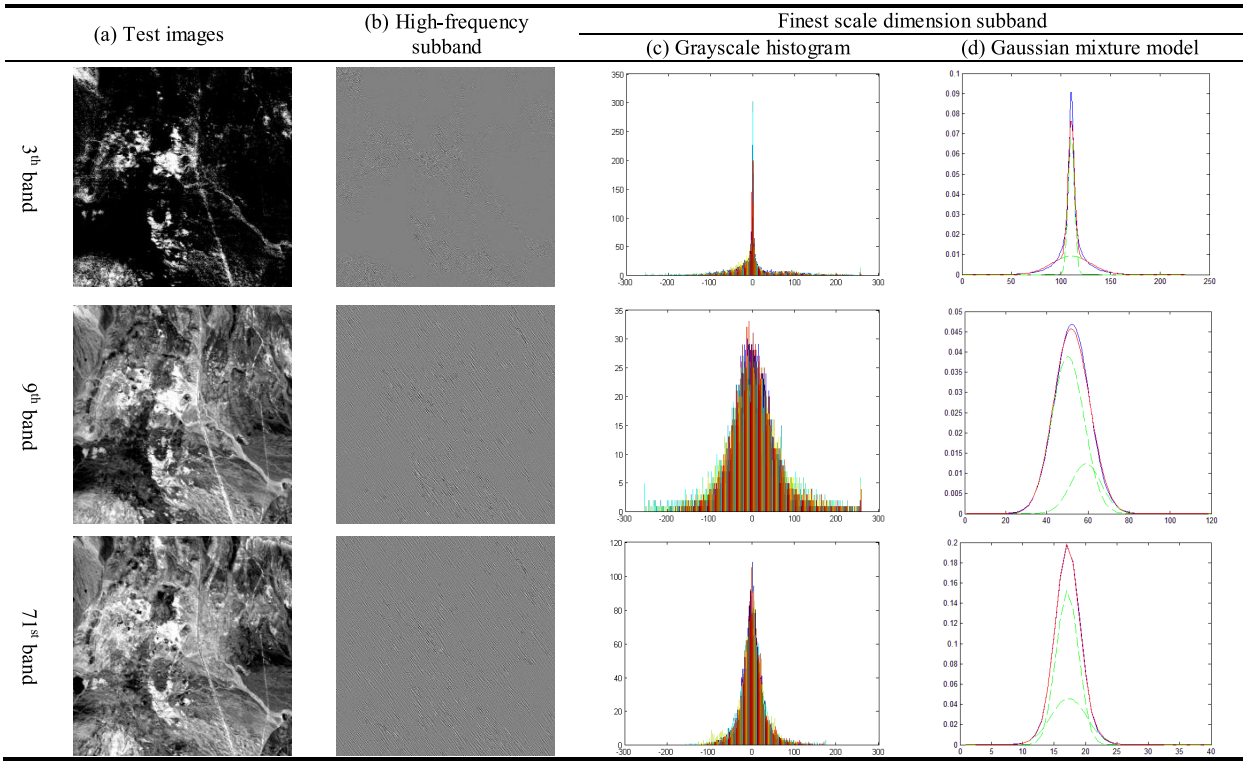


Fig. 2. Marginal statistical analysis of NSST coefficients.

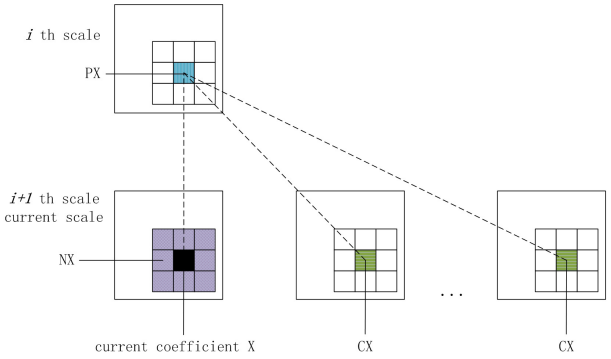


Fig. 3. Generalized neighborhood relationship of the NSST coefficients in the spatial direction.

The chi-plot is a scatter plot of (λ_i, χ_i). The greater deviation of the scatter point from the tolerance range (shaded gray area in Fig. 4) indicates a stronger correlation between the two data sets. By observing the chi-plot results shown in Fig. 4, we can see that the NSST coefficient shows a stronger correlation with its parent coefficient and neighborhood coefficients than with the cousin coefficients.

3) Multivariate Frequency Histogram: The multivariate frequency histogram is also used as a correlation evaluation method for evaluating the correlation between NSST coefficients with their generalized coefficients. It can be seen that the multivariate frequency histogram shows a characteristic of central symmetry, i.e., the NSST coefficients show a distribution trend that is similar to that of the generalized neighborhood coefficients.

4) Mutual Information: To evaluate the correlation objectively, the mutual information between X with its generalized neighborhood coefficients is calculated according to (4) [37]. By analyzing the results, the mutual information satisfies: $I(X, PX) > I(X, NX_i) > I(X, CX_j)$, where $i, j \in \{1, 2, \ldots, B\}$, B is the number of cousin coefficients of X

$$I(X, Y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dxdy. \tag{4}$$

From the experimental results, we can draw the following conclusions: 1) all points tend to the center; 2) the reference coefficient shows a certain correlation with its parent coefficient, neighborhood coefficient, and cousin coefficient; and 3) among the coefficients, the parent coefficient has a stronger correlation with the son coefficient.

According to the above analysis, the "parent–son" relationship is selected as the transfer relationship for the hidden Markov structure.

$$r_{ij} = \frac{\sum_{x=1}^{M} \sum_{y=1}^{N} ((I_i(x, y) - \overline{I}_i)(I_j(x, y) - \overline{I}_j))}{\sqrt{\sum_{x=1}^{M} \sum_{y=1}^{N} (I_i(x, y) - \overline{I}_i)^2 \sum_{x=1}^{M} \sum_{y=1}^{N} (I_j(x, y) - \overline{I}_j)^2}} \tag{5}$$

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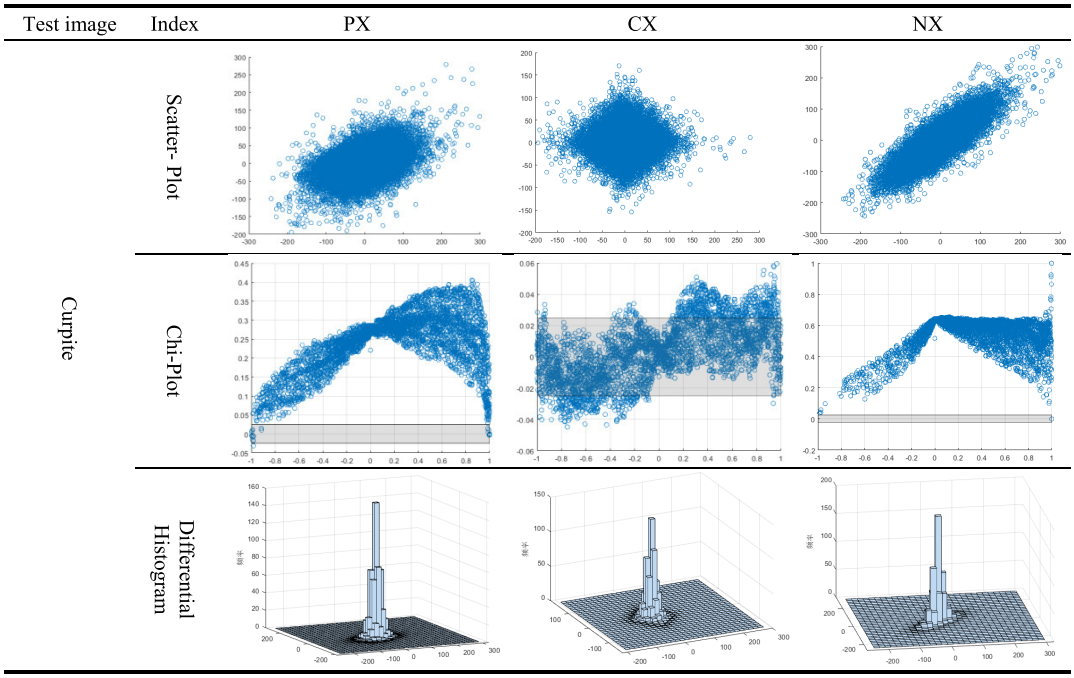


Fig. 4. Joint probability distribution of the direction subband coefficients of the NSST.

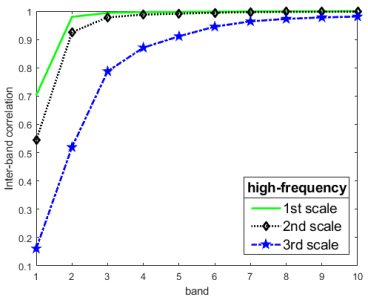


Fig. 5. Correlation statistics of subbands in the same scale and in the same direction of adjacent bands in the NSST band group.

D. Joint Statistical Analysis of NSST Coefficients in the Spectral Direction

The HS image reflects the reflection intensity of different electromagnetic bands of the object, which leads to a strong correlation among the HS spectra. To further analyze the correlation of the HS image NSST coefficients in the spectral direction, the band-group is defined according to the interspectral correlation of the HS images, and the correlation between adjacent bands and NSST coefficients for each band group is calculated as follows (5) as shown at the bottom of the previous page, where $I_i(x, y)$ and $I_j(x, y)$ are the ith and jth band images of the HS image with a size of $M \times N$, $\overline{I}_i = 1/(M \times N) \sum_{x=1}^{M} \sum_{y=1}^{N} I_i(x, y)$, $\overline{I}_j = 1/(M \times N) \sum_{x=1}^{M} \sum_{y=1}^{N} I_j(x, y)$, and r_{ij} is the correlation coefficient (CC) between $I_i(x, y)$ with $I_j(x, y)$.

Fig. 5 shows the correlation statistics of the high-frequency subbands corresponding to the adjacent bands of the band groups made of the first to the tenth bands of the HS image Cuprite. The horizontal axis shown in Fig. 5 represents the band number and the vertical axis is the correlation between the current band with the same scale and the same direction subband of the subsequent adjacent band. It can be seen that after the HS image is decomposed by NSST, the high-frequency subbands of the same scale and same direction in the current band and the subsequent adjacent bands exhibit a similar correlation property with the original HS images, although with the increase in the decomposition scale, the correlation is somewhat lower. However, it still shows a strong correlation, and the joint relationship between them show strong "clustering" and "aggregation" properties.

E. Analysis of Joint Correlation of Spatial–Spectral Directions of HS Image NSST Coefficients

From the previous study, it can be seen that there exists a strong correlation among HS image NSST coefficients in the band group in both the spatial and the spectral directions. Furthermore, the joint correlation in the two directions is studied, and the statistical results for the scatter plot are shown in Fig. 6. Fig. 6(a) and (b) present the statistical results of the joint correlation of the first band, where the subbands in the spatial direction of the first band are chosen as the spatial spread direction, and the correlation with their corresponding subbands in the second and third bands are studied. The detailed illustration of the specific subband joint correlation statistical spread path is depicted in Fig. 7. We define the ith band of the same-scale subband in the same direction as the $i + 1$th parent coefficient of the spectral direction. Fig. 6(c) shows the statistical result of the joint correlation between the second band as the spatial dimension and the subband corresponding to the third band. It can be seen from the statistical results that the HS image NSST direction subband spatial–spectral dimension joint correlation distribution has strong "clustering" and "aggregation" properties, indicating that the NSST coefficients have strong prediction ability in both the spatial and spectral directions.

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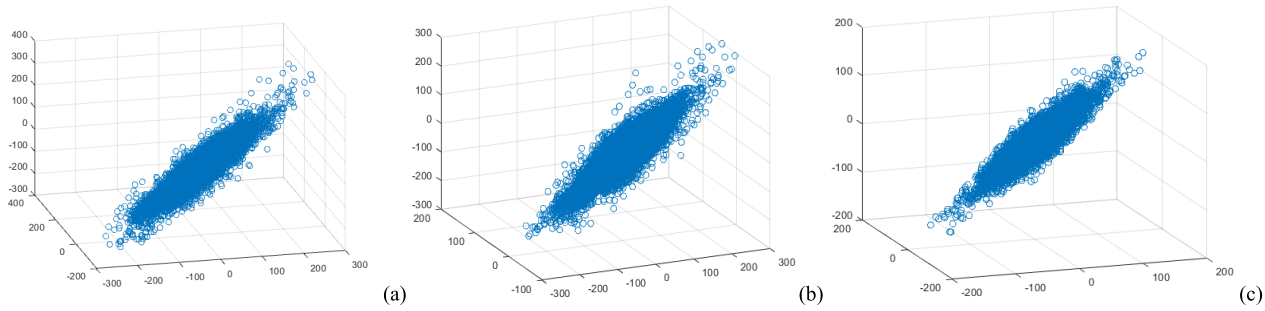


Fig. 6. Joint statistical distribution map of space–spectral dimension NSST high-frequency direction subbands in band group. (a) Statistical result of the joint correlation between the first band first scale as the spatial dimension and the subband corresponding to the second band first scale. (b) Statistical result of the joint correlation between the first band first scale as the spatial dimension and the subband corresponding to the third band first scale. (c) Statistical result of the joint correlation between the second band first scale as the spatial dimension and the subband corresponding to the third band first scale.

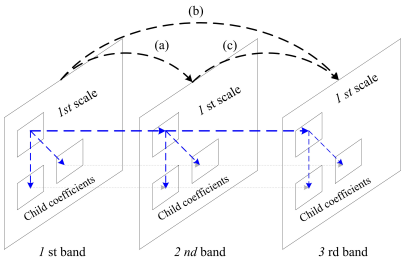


Fig. 7. Illustration diagram of joint correlation statistical path diagram of space–spectral dimension high-frequency direction subband.

Based on the data presented above, we can draw the following conclusions.

1) The NSST coefficients of the HS image and the PAN image exhibit the "sharp peak and heavy tail" property, which are well modeled by the GMM.
2) The NSST coefficients show strong "clustering" and "aggregation" properties in the spatial direction, which indicates the strong correlation of the NSST coefficient with its generalized neighborhood coefficients. At the same time, the parent coefficient exhibits the strongest prediction ability among the generalized neighborhood relationship.
3) Similar results can also be found in the spectral direction, so that the NSST direction subband coefficient of the current band in the HS image can be used to predict the subband coefficients of the corresponding direction of the latter bands.

III. HS Image NSST HMF

A. Nonsubsampled Shearlet Transform Hidden Markov Forest

Based on the previous conclusions, this article proposes a HS image NSST-HMF model that depicts the correlation of HS image NSST coefficients in both spatial and spectral directions, and it investigates the multidimensional coordinated transition among them. The state transition relationship of the proposed NSST-HMF is shown in Fig. 8.

The specific modeling process is introduced below.

H is the HS image band group with a size of $M \times N \times B$, where $M \times N$ is the size of the band image, and B is the band amount of the band group. First, each band image $\boldsymbol{H}_b (b \in \{1, 2, \ldots, B\})$ is decomposed into subbands by an L scale NSST transform, and each scale $j (j \in \{1, 2, \ldots, L\})$ has d_j directions. The tree structure formed by the coefficients of each direction $k (k \in \{1, 2, \ldots, d_j\})$ is denoted as $T^k_{b,j}$, where the subscript denotes the subtree root node j of the bth band, and $T^k_{b,1}$ represents the whole tree. $\boldsymbol{N}^k_b = \{N^k_{b,1}, N^k_{b,2}, \ldots, N^k_{b,p}\}$ represents the coefficient nodes in the $T^k_{b,1}$ and their corresponding hidden states, the parent coefficient of the spatial dimension corresponding to the coefficient $N^k_{b,i} (i \in \{2, 3, \ldots, p\})$ in the $T^k_{b,1}$ is $\rho^k_b(i)$, and the parent coefficient of the spectral dimension $N^k_{b,i} (i \in \{1, 2, \ldots, p\}, b \in \{2, 3, \ldots, n\})$ is $N^k_{b-1,i}$ (see the "red dot" in Fig. 8). Furthermore, based on the discussion in Section II-B, the GMM is used as the probability density distribution to model the NSST coefficients of each subband, and the state transition probability of the spatial and spectral directions are used to depict the transmission relationship among NSST coefficients with their parent coefficient of the two directions, respectively. Thus, the HMF Model $\theta_{\text{NSST-HMF}}$ that depicts the multidimensional synergistic correlation of the HS image NSST coefficients is established

$$\theta_{\text{NSST-HMF}} = \begin{cases} \left(P^m_{N^k_{b,1}},\ P^m_{N^k_{b-1,i}}, \varepsilon^{m,n}_{N^k_{b,i},\rho^k_b(i)},\ \varepsilon^{m,n}_{N^k_{b,i},N^k_{b-1,i}} \sigma^2_{N^k_{b,i},m},\ \mu_{N^k_{b,i},m}\right) \\ b = 2, \ldots, B; \quad i = 2, 3, \ldots, p; \quad m, n = 0, 1. \end{cases} \tag{6}$$

The specific meanings of the parameters are as follows.

1) $P^m_{N^k_{b,1}}$ is the state probability of the b band and k direction subband root $N^k_{b,1}$, and $P^m_{N^k_{b-1,i}}$ is the state probability of the spectral dimension parent coefficient $N^k_{b-1,i}$ of the b band and k direction subband coefficient $N^k_{b,i}$, where m is the hidden state, and m takes the values 1 and 0, indicating the large and small states, respectively. The meaning of n is the same.
2) $\varepsilon^{m,n}_{N^k_{b,i},\rho^k_b(i)}$ is a state transition probability in the spatial direction, and its expression is $\varepsilon^{m,n}_{N^k_{b,i},\rho^k_b(i)} = f(S^k_{b,N^k_{b,i}} = m, n | S^k_{b,\rho^k_b(i)} = m, n)$, which indicates that for the bth band and kth direction subband, when the hidden state $S^k_{b,\rho^k_b(i)}$ of the parent node $\rho^k_b(i)$ is m/n,

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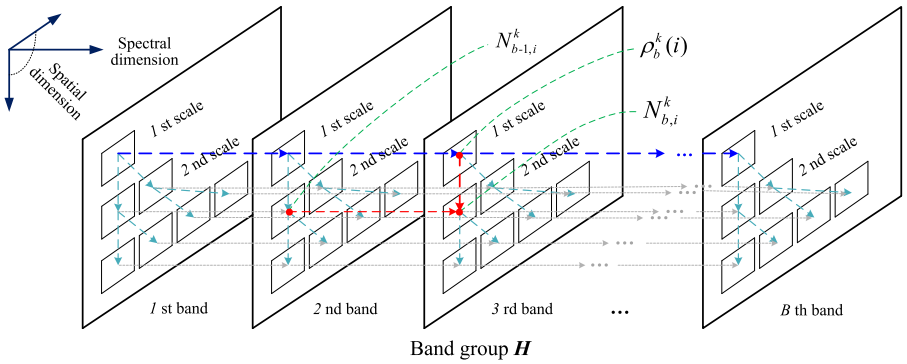


Fig. 8. Illustration of NSST-HMF multidimensional coefficient transfer relationship.

the conditional probability of the hidden state $S^k_{b,N^k_{b,i}}$ of the son node $N^k_{b,i}$ is m/n.

3) $\varepsilon^{m,n}_{N^k_{b,i},N^k_{b-1,i}}$ is a state transition probability in the spectral direction, and its expression is

$$\varepsilon^{m,n}_{N^k_{b,i},N^k_{b-1,i}} = f\left(S^k_{b,N^k_{b,i}} = m, n \,|\, S^k_{b,N^k_{b-1,i}} = m, n\right)$$

which means that for the bth band and kth direction subband, the conditional probability of the hidden state $S^k_{b,N^k_{b,i}}$ of the son node $N^k_{b,i}$ is m/n.

4) $\sigma^2_{N^k_{b,i},m}$ and $\mu_{N^k_{b,i},m}$ are the variance and mean of the bth band and kth direction subband at the $N^k_{b,i}$ node spatially hidden state $S^k_{b,i}$, respectively.

B. Parameter Estimation of the NSST-HMF Model

The EM algorithm is used to estimate the model parameters involving the spatial direction [38]. To estimate the parameters involving the spectral direction, a classification [39] based on the fast parameter estimation method is proposed.

1) Fast Classification Method for HS Image NSST Coefficients in the Band Group: In the HS band group H, each band image $\boldsymbol{H}_b (b \in \{1, 2, \ldots, B\})$ is decomposed into subbands by NSST. The following equation gives the classification discriminant method for the large and small coefficients for the k direction subband coefficients [39]:

$$S^k_b(\cdot) = \begin{cases} 1, & \text{if } |\cdot|/\sigma_{b,k} \geq \mathrm{T}_{b,k} \\ 0, & \text{else} \end{cases} \tag{7}$$

where $\sigma_{b,k}$ is the standard deviation of the b band and k direction subband coefficient, $\mathrm{T}_{b,k}$ is the given threshold (the experimental statistics can be selected from 2.3 to 2.5, 2.4 is used in the experiment), and $S^k_b(\cdot)$ is 1 or 0, representing the current coefficient of a large coefficient or a small coefficient, respectively. For the current subband coefficient $N^k_{b,i}$ and its parent coefficient $N^k_{b-1,i}$ of the spectral direction, $S^k_b(N^k_{b,i})$ and $S^k_{b-1}(N^k_{b-1,i})$ are the classified-value bands obtained according to (6).

2) Parameter Estimation Method for Spectral Direction Parameters: D^k_b is a subband of the band group H and its corresponding parent subband D^k_{b-1} of the spectral direction is classified into the binary direction subbands $D_B^k_b$, $D_B^k_{b-1}$ by (6). The parameter estimation process for the spectral direction parameters is introduced below.

1) State probability $P^m_{N^k_{b-1,i}}$ of parent coefficient $N^k_{b-1,i}$

$$P^m_{N^k_{b-1,i}} = \frac{\text{number}\left(\boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m}\right)}{\left\langle \boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m} \right\rangle} \tag{8}$$

where $\boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m}$ indicates a set of locations in $D_B^k_{b-1}$ whose state values in the neighborhood centered on $N^k_{b-1,i}$ corresponding location are m (select the 3×3 neighborhood centered on $N^k_{b-1,i}$ in this experiment), $< \boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m} >$ is the size of the neighborhood coefficients where $\boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m}$ is located, and number($\boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m}$) is the size of $\boldsymbol{L}_{D_B^k_{b-1},N^k_{b-1,i},m}$.

1) State transition probability $\varepsilon^{m,n}_{N^k_{b,i},N^k_{b-1,i}}$

$$\varepsilon^{m,n}_{N^k_{b,i},N^k_{b-1,i}} = \frac{\text{number}\left(V_{D_B^k_b,N^k_{b,i}} = m \,\&\, V_{D_B^k_{b-1},N^k_{b-1,i}} = n\right)}{\text{number}\left(V_{D_B^k_{b-1},N^k_{b-1,i}} = n\right)} \tag{9}$$

where $V_{D_B^k_b,N^k_{b,i}}$ represents the value of the corresponding $N^k_{b,i}$ and the position in $D_B^k_b$, and $V_{D_B^k_{b-1},N^k_{b-1,i}}$ is the value of the corresponding $N^k_{b-1,i}$ position in the $D_B^k_{b-1}$.

Table I shows the NSST-HMF model parameters of the band group that contains the ninth and tenth bands of the HS image cuprite. In Table I, P is the initial probability, A is the state probability transition matrix in the spatial direction, σ is the variance of Gaussian mixture distribution, S is the state probability transition matrix in the spectral direction, and k is the hidden state.

IV. HS-Pansharpening Algorithm Based on NSST-HMF

Based on the above NSST-HMF model, this article proposed a HS-pansharpening algorithm. The flowchart of the proposed algorithm is shown in Fig. 9.

A. Spectral Modulation of PAN Images

To improve the spectral coverage of the PAN image to match the wavelength acquisition range of the HS bands in

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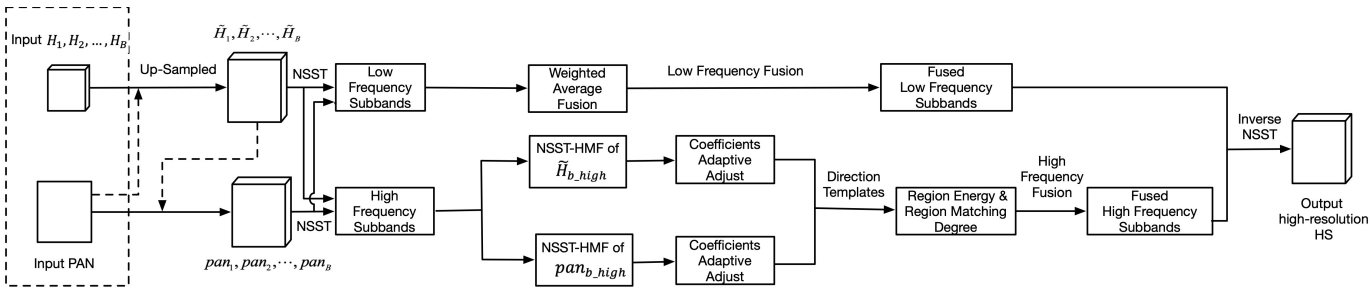


Fig. 9. Flowchart of the fusion algorithm in this article.

TABLE I
NSST-HMF MODEL PARAMETERS OF THE 9TH BAND OF THE HS IMAGE CUPRITE

		K=1		K=2	
		State1	State2	State1	State2
$P_{A_1,k}$		0.6919	0.3081	0.6518	0.3482
$A_{2,2k-1}$	State1	0.0277	0.8065	0.9857	0.0797
	State2	0.9723	0.1935	0.0143	0.9203
$A_{2,2k}$	State1	0.9501	0.2700	0.0160	0.9237
	State2	0.0499	0.7300	0.9840	0.0763
$A_{3,2k-1}$	State1	0.0134	0.9959	0.9739	0.6363
	State2	0.9866	0.0041	0.0261	0.3637
$A_{3,2k}$	State1	0.9508	0.0005	0.0238	0.3638
	State2	0.0492	0.9995	0.9762	0.6362
$A_{3,2k+3}$	State1	0.0000	0.9875	0.4585	0.0311
	State2	1.0000	0.0125	0.5415	0.9689
$A_{3,2k+4}$	State1	1.0000	0.0143	0.5099	0.9626
	State2	0.0000	0.9857	0.4901	0.0374
$\sigma_{1,k}$		4.1105	8.4717	4.5345	8.6078
$\sigma_{2,2k-1}$		5.6516	2.6353	2.3144	5.4416
$\sigma_{2,k}$		2.6921	5.6450	5.4321	2.3105
$\sigma_{3,2k-1}$		1.2651	3.0190	1.1523	3.3387
$\sigma_{3,2k}$		3.0693	1.2725	3.3384	1.1599
$\sigma_{3,2k+3}$		3.1794	1.1302	3.0599	1.2168
$\sigma_{2,2k+4}$		1.1305	3.1816	1.1956	2.9729
$P_{S_1,k}$		0.4990	0.5010	0.5038	0.4962
$S_{2,2k-1}$	State1	0.8820	0.1178	0.8476	0.1541
	State2	0.1180	0.8822	0.1524	0.8459
$S_{2,2k}$	State1	0.8635	0.1364	0.8619	0.1390
	State2	0.1365	0.8636	0.1381	0.8610
$S_{3,2k-1}$	State1	0.7589	0.2418	0.6917	0.3102
	State2	0.2411	0.7582	0.3083	0.6898
$S_{3,2k}$	State1	0.8043	0.1965	0.7283	0.2730
	State2	0.1957	0.8035	0.2717	0.7270
$S_{3,2k+3}$	State1	0.7824	0.2178	0.7429	0.2574
	State2	0.2176	0.7822	0.2571	0.7426
$S_{3,2k+4}$	State1	0.7047	0.2961	0.7300	0.2707
	State2	0.2953	0.7039	0.2700	0.7293

HS-pansharpening, the single spatial detail (SSD) model [40] is used to perform adaptive spectral modulation for the PAN image based on HS image bands. The specific process is as follows.

By upsampling the b band image $H_b(b \in \{1, 2, \ldots, B\})$ of the HS image band group, the corresponding band image $\tilde{H}_b$ is obtained. The PAN image is further spectrally modulated based on $\tilde{H}_b$ to obtain the following modulated PAN image pan_b:

$$\text{pan}_b = \gamma_b D \tag{10}$$

where D is the detailed image extracted from the PAN by the zero-phase filter, and γ_b is calculated by using the minimum variance unbiased (MVU) estimator

$$\gamma_b = \frac{D_R^{\mathrm{T}}(\tilde{H}_b - \tilde{H}_{bR})}{||D_R||} \tag{11}$$

where R denotes the spatially degraded data. By assuming that the model gain parameters γ_b are approximately constant across scales, in this experiment, 5, 6, 4 are respectively selected for the three data sets.

B. Fusion Rule of the Low-Frequency Subband

The weighted average fusion scheme based on the correlation of generalized neighborhood coefficients is used to fuse the low-frequency subbands $\tilde{H}_{b_\text{low}}$ with $\text{pan}_{b_\text{low}}$. The specific process is demonstrated below.

Let $c_{\tilde{H}_b_\text{low}}(i, j)$ and $c_{\text{pan}_b_\text{low}}(i, j)$ be the coefficients of $\tilde{H}_{b_low}$ and $\text{pan}_{b_\text{low}}$ at the position (i, j), respectively. The mean $\mu_{\text{HP}_\text{low}}(i, j)$ and regional variance $SD_{\text{HP}_\text{low}}(i, j)$(HP_low $\in \{\tilde{H}_b_\text{low}$,pan_$b$_low$\}$) in the 3×3 region centered on $c_{\tilde{H}_b_\text{low}}(i, j)$ and $c_{\text{pan}_b_\text{low}}(i, j)$ are calculated as follows.

$$\begin{cases} \mu_{\text{HP}_\text{low}}(i, j) = \dfrac{1}{3 \times 3} \displaystyle\sum_{m=-1}^{1} \sum_{n=-1}^{1} c_{\text{HP}_\text{low}}(i+m, j+n) \\ \text{SD}_{\text{HP}_\text{low}}(i, j) \\ \quad = \sqrt{\dfrac{1}{3 \times 3} \displaystyle\sum_{m=-1}^{1} \sum_{n=-1}^{1} \left(\begin{matrix} c_{\text{HP}_\text{low}}(i+m, j+n) \\ -\mu_{\text{HP}_\text{low}}(i, j) \end{matrix} \right)^2}. \end{cases} \tag{12}$$

The regional variance indicates the deviation of the coefficient from the regional average coefficient value. The larger value indicates a greater deviation, which also indicates the rich texture. Therefore, more attention should be paid to the coefficient during the fusion process, i.e., the weight of the coefficient should be larger. According to this, the following equation gives the fusion rule of low-frequency subband $\tilde{H}_{b_\text{low}}$ with $\text{pan}_{b_\text{low}}$:

$$\begin{aligned} &c_{\tilde{H}_b_\text{low}\&\text{pan}_b_\text{low}}(i, j) \\ &\quad = \omega_{\tilde{H}_b_\text{low}}(i, j) \times c_{\tilde{H}_b_\text{low}}(i, j) \\ &\qquad + \omega_{\text{pan}_b_\text{low}}(i, j) \times c_{\text{pan}_b_\text{low}}(i, j) \end{aligned} \tag{13}$$

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where $c_{\tilde{H}_b_\text{low}\&\text{pan}_b_\text{low}}$ is the fused coefficient of the low-frequency subband $\tilde{H}_{b_\text{low}}$ and $\text{pan}_{b_\text{low}}$, and $\omega_{\tilde{H}_b_\text{low}}$ and $\omega_{\text{pan}_b_\text{low}}$ are the weights, which can be calculated as follows:

$$\begin{cases} \omega_{\tilde{H}_b_\text{low}}(i,j) = \dfrac{\text{SD}_{\tilde{H}_b_\text{low}}(i,j)}{\text{SD}_{\tilde{H}_b_\text{low}}(i,j)+\text{SD}_{\text{pan}_b_\text{low}}(i,j)} \\ \omega_{\tilde{H}_b_\text{low}}(i,j) = \dfrac{\text{SD}_{\text{pan}_b_\text{low}}(i,j)}{\text{SD}_{\tilde{H}_b_\text{low}}(i,j)+\text{SD}_{\text{pan}_b_\text{low}}(i,j)} \end{cases} \tag{14}$$

C. Fusion Rule of the High-Frequency Subbands

For the high-frequency subband, by training the HMF model, the NSST-HMF model parameter sets $\theta_{\text{NSST-HMF},\,\tilde{H}_b_\text{high}}$ and $\theta_{\text{NSST-HMF},\,\text{pan}_b_\text{high}}$ of $\tilde{H}_{b_\text{high}}$ and $\text{pan}_{b_\text{high}}$ are obtained according to (15), as shown at the bottom of this page, where $b = 2,\ldots,B$; $i = 2,3,\ldots,p$; $m,n = 0,1$, and k is the direction of the subband.

To utilize the transfer relationship of the high-frequency subbands, the spatial–spectral joint correlation of the state probabilities $p^m_{N^k_{b,i,\boldsymbol{HP_high}},A\&S}$ of the reference coefficients $N^k_{b,i,\boldsymbol{HP_high}}(\boldsymbol{HP_high} \in \{\tilde{H}_b_\text{high}, \text{pan}_b_\text{high}\})$ is defined to adjust the NSST coefficients

$$\begin{cases} p^1_{N^k_{b,i,\boldsymbol{HP_high}},A\&S} = \dfrac{p^1_{N^k_{b,i,\boldsymbol{HP_high}},A} + p^1_{N^k_{b,i,\boldsymbol{HP_high}},S}}{1+\xi_{\boldsymbol{HP_high}}} \\ p^0_{N^k_{b,i,\boldsymbol{HP_high}},A\&S} = 1 - p^1_{N^k_{b,i,\boldsymbol{HP_high}},A\&S} \end{cases} \tag{16}$$

where $\xi_{\boldsymbol{HP_high}}$ is the spectral CC. $p^m_{N^k_{b,i,\boldsymbol{HP_high}},A}$, $p^m_{N^k_{b,i,\boldsymbol{HP_high}},S}$ are the spatial and spectral state probabilities of $\tilde{H}_{b_\text{high}}$ and $\text{pan}_{b_\text{high}}$, respectively, based on the spatial–spectral state transition probabilities

$$\begin{cases} p^m_{N^k_{b,1,\boldsymbol{HP_high}},A} = P^m_{N^k_{b,1,\boldsymbol{HP_high}}} \\ p^1_{N^k_{b,i,\boldsymbol{HP_high}},A} \\ \quad = \begin{pmatrix} p^1_{N^k_{b,i-1,\boldsymbol{HP_high}},A} \times \varepsilon^{1,1}_{N^k_{b,i,\boldsymbol{HP_high}},\rho^k_{b,\boldsymbol{HP_high}}(i)} \\ +p^0_{N^k_{b,i-1,\boldsymbol{HP_high}},A} \times \varepsilon^{0,1}_{N^k_{b,i,\boldsymbol{HP_high}},\rho^k_{b,\boldsymbol{HP_high}}(i)} \end{pmatrix} \\ p^0_{N^k_{b,i,\boldsymbol{HP_high}},A} = 1 - p^1_{N^k_{b,i,\boldsymbol{HP_high}},A} \end{cases} \tag{17}$$

$$p^1_{N^k_{b,i,\boldsymbol{HP_high}},\mathbf{S}} = \xi_{\boldsymbol{HP_high}} \begin{pmatrix} P^1_{N^k_{b-1,i,\boldsymbol{HP_high}}} \times \varepsilon^{1,1}_{N^k_{b,i,\boldsymbol{HP_high}},N^k_{b-1,i,\boldsymbol{HP_high}}} \\ +P^0_{N^k_{b-1,i,\boldsymbol{HP_high}}} \times \varepsilon^{0,1}_{N^k_{b,i,\boldsymbol{HP_high}},N^k_{b-1,i,\boldsymbol{HP_high}}} \end{pmatrix}. \tag{18}$$

Based on the joint correlation of the state probability $p^m_{N^k_{b,i,\boldsymbol{HP_high}},A\&\,\mathrm{S}}$, the expected state probability of $N^k_{b,i,\boldsymbol{HP_high}}$ is defined as follows, where the probability density function value is calculated by (20):

$$p\left(m|N^k_{b,i,\boldsymbol{HP_high}}\right) = \frac{\dfrac{p^m_{N^k_{b,i,\boldsymbol{HP_high}},A\&\mathbf{S}}}{\sqrt{2\pi\sigma^2_{N^k_{b,i,\boldsymbol{HP_high}},m}}} \exp\left(-\dfrac{\left(N^k_{b,i,\boldsymbol{HP_high}} - \mu_{N^k_{b,i,\boldsymbol{HP_high}},m}\right)^2}{2\sigma^2_{N^k_{b,i,\boldsymbol{HP_high}},m}}\right)}{f\left(N^k_{b,i,\boldsymbol{HP_high}}\right)}. \tag{19}$$

$$\begin{aligned} f\left(N^k_{b,i,\boldsymbol{HP_high}}\right) &= \frac{p^1_{N^k_{b,i,\boldsymbol{HP_high}},A\&\mathbf{S}}}{\sqrt{2\pi\sigma^2_{N^k_{b,i,\boldsymbol{HP_high}},1}}} \times \exp\left(-\frac{\left(N^k_{b,i,\boldsymbol{HP_high}} - \mu_{N^k_{b,i,\boldsymbol{HP_high}},1}\right)^2}{2\sigma^2_{N^k_{b,i,\boldsymbol{HP_high}},1}}\right) \\ &\quad + \frac{p^0_{N^k_{b,i,\boldsymbol{HP_high}},A\&\mathbf{S}}}{\sqrt{2\pi\sigma^2_{N^k_{b,i,\boldsymbol{HP_high}},0}}} \times \exp\left(-\frac{\left(N^k_{b,i,\boldsymbol{HP_high}} - \mu_{N^k_{b,i,\boldsymbol{HP_high}},0}\right)^2}{2\sigma^2_{N^k_{b,i,\boldsymbol{HP_high}},0}}\right) \end{aligned} \tag{20}$$

Meanwhile, eight 3×3 direction templates $\boldsymbol{D}_s (s = 1,2,\ldots,8)$ [41] are defined in Fig. 10 to depict the directional information of the images. According to the direction feature of $N^k_{b,i,\boldsymbol{HP_high}}$, the corresponding direction template $\boldsymbol{D}_r (r \in \{1,2,\ldots,8\})$ is selected from $\boldsymbol{D}_s$, based on the state expectation probability of the coefficient in the direction region in the $\boldsymbol{D}_r$ direction, and further determining the probability that the large state coefficient of the direction region occupies a large state coefficient in the entire region, which is denoted as $p_{\boldsymbol{D}_r}(N^k_{b,i,\boldsymbol{HP_high}})$, then adjusting the coefficient $N^k_{b,i,\boldsymbol{HP_high}}$ to $\hat{N}^k_{b,i,\boldsymbol{HP_high}}$ based on $p_{\boldsymbol{D}_r}(N^k_{b,i,\boldsymbol{HP_high}})$ through (21)

$$\hat{N}^k_{b,i,\boldsymbol{HP_high}}(x,y) = \frac{p_{\boldsymbol{D}_r}\left(N^k_{b,i,\boldsymbol{HP_high}}(x,y)\right)}{N^k_{b,i,\boldsymbol{HP_high}}(x,y)^{-1}} \tag{21}$$

$$p_{\boldsymbol{D}_r}\left(N^k_{b,i,\boldsymbol{HP_high}}(x,y)\right) = \frac{p\left(m=1\middle|N^k_{b,i,\boldsymbol{HP_high}}(x,y)\right)}{\sum_{x=1}^{3}\sum_{y=1}^{3} p\left(m=1\middle|N^k_{b,i,\boldsymbol{HP_high}}(x,y)\right)} \tag{22}$$

where $x = 1,2,3$ and $y = 1,2,3$.

$$\begin{cases} \theta_{\text{NSST}-\text{HMF},\,\tilde{H}_b_\text{high}} \\ = \left\{ \left(P^m_{N^k_{b,1,\,\tilde{H}_b_\text{high}}},\ P^m_{N^k_{b-1,i,\,\tilde{H}_b_\text{high}}},\varepsilon^{m,n}_{N^k_{b,i,\,\tilde{H}_b_\text{high}},\rho^k_{b,\,\tilde{H}_b_high}(i)},\ \varepsilon^{m,n}_{N^k_{b,i,\,\tilde{H}_b_\text{high}},N^k_{b-1,i,\,\tilde{H}_b_\text{high}}},\sigma^2_{N^k_{b,i,\,\tilde{H}_b_\text{high}},m},\ \mu_{N^k_{b,i,\,\tilde{H}_b_\text{high}},m} \right) \right\} \\ \theta_{\text{NSST}-\text{HMF},\,\text{pan}_b_\text{high}} \\ = \left\{ \left(P^m_{N^k_{b,1,\text{pan}_b_\text{high}}},\ P^m_{N^k_{b-1,i,\text{pan}_b_\text{high}}},\varepsilon^{m,n}_{N^k_{b,i,\text{pan}_b_\text{high}},\rho^k_{b,\text{pan}_b_\text{high}}(i)},\sigma^2_{N^k_{b,i,\text{pan}_b_\text{high}},m},\ \mu_{N^k_{b,i,\text{pan}_b_\text{high}},m} \right) \right\} \end{cases} \tag{15}$$

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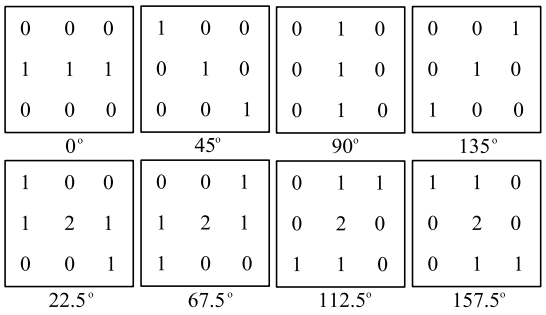


Fig. 10. Eight-direction templates $\boldsymbol{D}_S$.

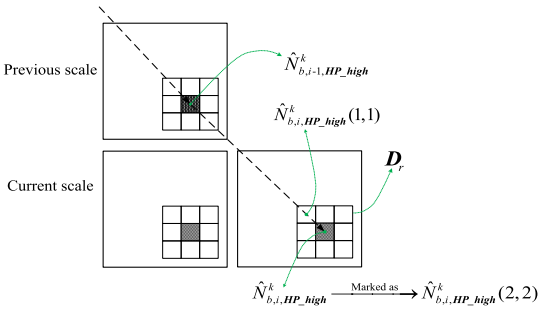


Fig. 11. Illustration diagram of the calculation of the matching of energy and direction regions in the direction area.

Then, $\hat{N}^k_{b,i,\boldsymbol{HP_high}}$ is used to calculate the direction region energy $E_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\boldsymbol{HP_high}}}$ and the direction region matching degree $M_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\tilde{H}_b_high},\hat{N}^k_{b,i,pan_b_high}}$ is based on $\boldsymbol{D}_r$. Fig. 11 shows the diagram of the calculation of the matching of energy and direction regions in the direction region

$$E_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\boldsymbol{HP_high}}} = \sum_{x=1}^{3}\sum_{y=1}^{3}\left(\boldsymbol{D}_r(x,y)\times \hat{N}^k_{b,i,\boldsymbol{HP_high}}(x,y)\right)^2 \tag{23}$$

$$M_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}} = \frac{2\sum_{x=1}^{3}\sum_{y=1}^{3}\left(\begin{array}{l}(\boldsymbol{D}_r(x,y))^2\times \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}}(x,y)\\ \times \hat{N}^k_{b,i,pan_b_\mathrm{high}}(x,y)\end{array}\right)}{E_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}}} + E_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}}. \tag{24}$$

Finally, the matching degree threshold $\beta_{\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}$ between $\mathrm{t}\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}}$ and $\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}$ is determined by the maximum interclass variance method (OTSU) [42], and the fusion rule of the high-frequency subbands [41] is defined as follows:

If $M_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}} > \beta_{\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}$, (25), as shown at the bottom of this page, is used to determine the fusion result, where

$$\begin{cases}\omega_{\min,\ \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}} \\ \quad = \dfrac{M_{D_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_high}} - \beta_{\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}}{2\left(1-\beta_{\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\right)} \\ \omega_{\max,\ \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}} \\ \quad = 1-\omega_{\min,\ \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\end{cases} \tag{26}$$

If $M_{\boldsymbol{D}_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}} \le \beta_{\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}$, we use the following equation to determine the fusion result:

$$\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}\,\&\,\mathrm{pan}_b_\mathrm{high}} = \begin{cases}\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}}, & \mathrm{if}\left(E_{D_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}}} > E_{D_r,\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\right)\\ \hat{N}^k_{b,i,\ \mathrm{pan}_b_\mathrm{high}}, & \mathrm{others}\end{cases} \tag{27}$$

V. Experiments and Analysis

A. Experimental Settings and Data Sets Description

In this article, three open data sets are used for the experiments. The first is the "Moffett field data set," which captures a mixed urban/rural scene of the Moffett region. The size of the PAN image is 180 × 180, whereas the size of the HS image is 36 × 36, which is a spatial resolution ratio of 5 between the two images. The HS image [11] was acquired by the airborne HS instrument AVIRIS which contains 176 bands in the spectral range from 0.4 to 2.5 μm. The second data set is the "Chikusei data set" [43], which captures the city/agricultural area of Ibaraki, Japan. The size of the PAN image is 420 × 420, whereas the size of the HS image is 70 × 70, and the spatial resolution ratio between the two images is 6. The airborne HS data set was taken using Headwall's HS Visible and Near-Infrared series C (VNIR-C) imaging sensor, which contains 128 bands in the spectral range from 0.363 to 1.018 μm. The third data set is the "University of Pavia data set" [7]. The size of the PAN image is 320 × 320, whereas the size of the HS image is 80 × 80, and the spatial resolution ratio between the two images is 4. The HS image was acquired by the ROSIS-3 optical airborne sensor and the 12 noisy bands have been removed, and a total of 103 bands

$$\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}\,\&\,\mathrm{pan}_b_\mathrm{high}} = \begin{cases}\omega_{\max,\ \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\times \hat{N}^k_{b,i,\ \tilde{H}_b_\mathrm{high}} + \omega_{\min,\ \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\times \hat{N}^k_{b,i,\ \mathrm{pan}_b_\mathrm{high}}, & \\ \qquad\qquad \mathrm{if}\left(E_{D_r,\hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}}} > E_{D_r,\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\right) & \\ \omega_{\min,\ \hat{N}^k_{b,i,\tilde{H}_b_high},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\times \hat{N}^k_{b,i,\ \tilde{H}_b_\mathrm{high}} + \omega_{\max,\ \hat{N}^k_{b,i,\tilde{H}_b_\mathrm{high}},\hat{N}^k_{b,i,\mathrm{pan}_b_\mathrm{high}}}\times \hat{N}^k_{b,i,\ \mathrm{pan}_b_\mathrm{high}}, & \\ \qquad\qquad \mathrm{others} & \end{cases} \tag{25}$$

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in the spectral range from 0.430 to 0.838 μm is used in the experiment.

To verify the effectiveness of the proposed method, we compared our model with the following HS-pansharpening methods.

1) *Gram–Schmidt (GS) [13]*: The Gram–Schmidt (GS) method is an improved CS pansharpening method, which computes the synthetic intensity component by performing a linear regression between a high-resolution image and lower-resolution bands to mitigate spectral distortion.
2) *Principal Component Analysis (PCA) [16], [45]*: In the principal component analysis (PCA) method the main spatial information of HS is replaced by the high spatial information contained in the PAN image to ensure the high fidelity in rendering the spatial details.
3) *Gram–Schmidt Adaptive (GSA) [13], [16]*: The Gram–Schmidt adaptive (GSA) method was proposed based on GS, where a synthetic low-resolution PAN image is generated by the linear model with weights estimated by the minimization of the mean square error between the estimated component and a filtered downsampled PAN image.
4) *Guided Filter Principal Component Analysis (GFPCA) [18]*: The guided filter principal component analysis (GFPCA) method uses the PAN image to guide the filtering process, and the method transfers the spatial structures of the high-resolution PAN image to the enhanced HS image.
5) *Coupled Nonnegative Matrix (CNMF) [21], [22]*: The coupled nonnegative matrix (CNMF) alternately unmixes both sources of data to obtain the endmember spectra and the high spatial resolution abundance maps to get the pansharpening results.
6) *Wang et al. [44]*: The proposed method in [44] is a PAN and MS image pansharpening algorithm based on an adaptive neural network and sparse representation in the NSST domain.
7) Wang *et al. [41]*: The proposed method in [41] is an NSST MS pansharpening method based on directional neighborhood matching, where a binary tree structure has been constructed to model the tree structure matching.

Parameter tuning and algorithm settings are performed as follows.

1) For all the methods that require NSST, the "maxflat" nonsubsampled multiscale filter and the improved shear filter bank were selected. And the subband directions in each scale are 2, 4, and 8, from coarser to finer.
2) In the proposed method, $T_{b,k}$ is set to 2.4 for the experiments.
3) *Wang et al.* [41] and [44] are designed for MS-pansharpening, the MS images are first transformed into their intensity-hue-saturation (IHS) domain, and the I component is decomposed into subbands using NSST transform. To apply these methods in HS-pansharpening, we use each band image to replace the I component in the experiment, and iteratively apply the pansharpening process of each band.
4) Experiments are conducted with Intel Core i7 6700 CPU 3.4 GHz with 32-GB RAM in a Windows 10 system.

B. Experimental Results

1) Subjective Visual Evaluation: Figs. 12–14 show the pansharpening results using different methods of the three experimental data sets with a local area enlarging 2.89 times. Fig. 12 shows the pseudocolor fusion result of the Moffett field data set, which is synthesized by the 121st, 148th, and 176th bands. Fig. 13 shows the pseudocolor fusion result from the Chikusei data set, which is synthesized by the 100th, 117th, and 128th bands, Fig. 14 shows the pseudocolor fusion result from the University of Pavia data set, which is synthesized by the 55th, 75th, and 25th bands.

From Figs. 12–14, we can observe that most pansharpening methods can achieve promising results except the GFPCA method. By observing Figs. 12(f) and 13(f), the fused image of GFPCA is significantly blurred, and this is because numerous detailed features appear in the Moffett field data set and Chikusei data set but for the data set of University of Pavia, in which there exist more smooth areas, the blur effect is slighter.

From a comparison of the color preservation with the reference image, the proposed method achieves superior performance in the compared methods. At the same time, the texture features of the fused image can be better maintained, which indicates that the spectral distortion is slighter and the spatial resolution of the HS image has been well improved.

From the local enlargement results, we can observe that the obvious ambiguity phenomenon appears at the boundaries of Fig. 14(h) of the GS method and Figs. 12(h) and 14(h) of the proposed method in [44]. The geometrical information changes after fusion as shown in Fig. 13(h). The GSA, PCA, and CNMF [41] methods achieve promising performance in-line with the geometrical details, but the strong bias of the color can be observed which indicates that strong spectral distortion occurs, and the spectral information of the HS images is not well preserved. The proposed method can well preserve the spatial details and the spectral information of the HS images simultaneously.

2) Objective Evaluation: Tables II–IV show the statistical results of objective evaluation indicators of different algorithms. The CC is a cross-correlation metric, which characterizes the geometric distortion, and the ideal value of CC is 1; the spectral angle mapper (SAM) is a spectral angular quantity index, which reveals the degree of spectral shape retention, and in our experiments, we use the average SAM value with respect to pixels for the quality index, and the ideal value of SAM is 0; Erreur Relative Global Adimensionnelle de Synthese (ERGAS) is an overall relative error metric that offers a global indication of the quality of a fused image, and the ideal value of ERGAS is 0; the root-mean-square error (RMSE) is used to measure the deviation between the pansharpening image and the reference image, which is a global measure, and the ideal value of RMSE is 0.

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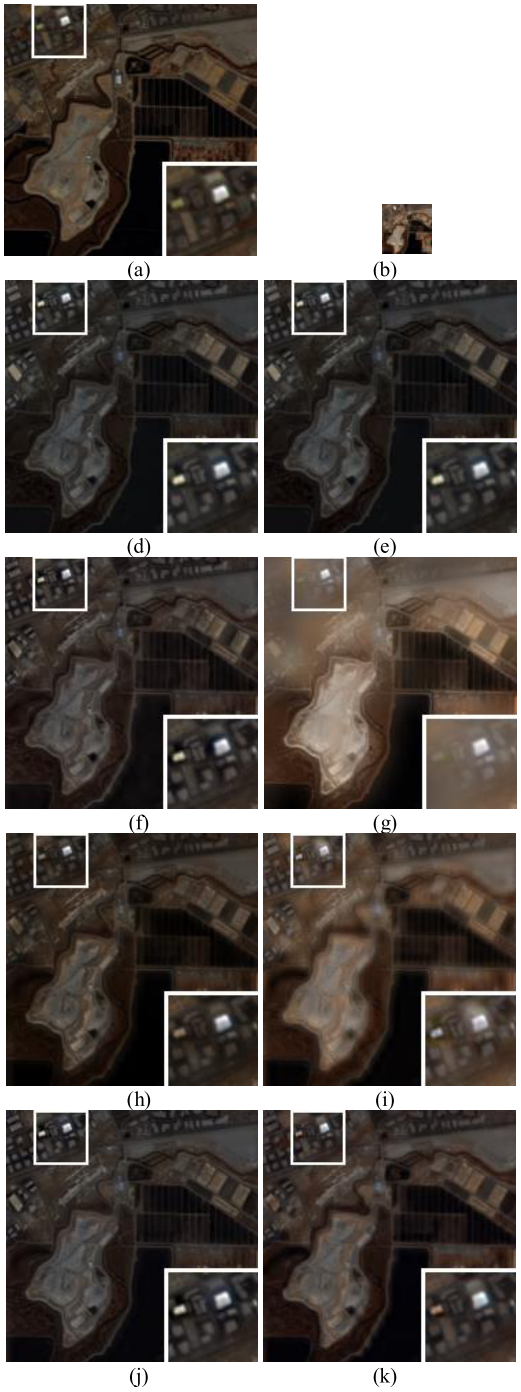


Fig. 12. Fusion results of the Moffett field data set. (a) Reference image. (b) HS image. (c) GS. (d) PCA. (e) GSA. (f) GFPCA. (g) CNMF. (h) Wang *et al.* [44]. (i) Wang *et al.* [41]. (j) NSST-HMF.

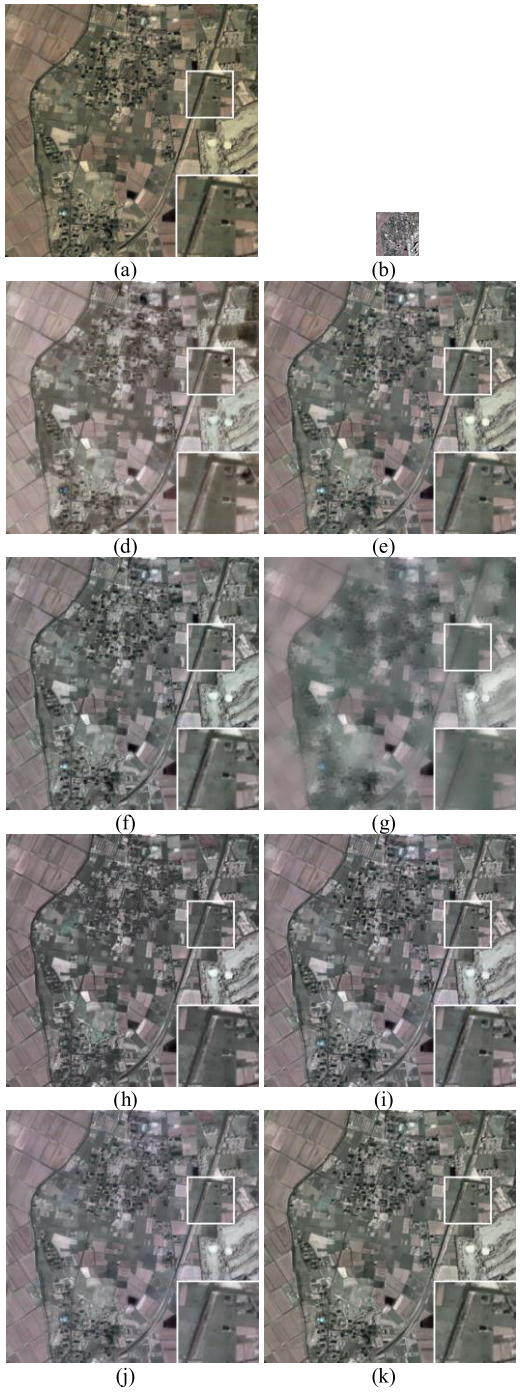


Fig. 13. Fusion results of the Chikusei data set. (a) Reference image. (b) HS image. (c) GS. (d) PCA. (e) GSA. (f) GFPCA. (g) CNMF. (h) Wang *et al.* [44]. (i) Wang *et al.* [41]. (j) NSST-HMF.

It can be seen from Tables II–IV that our method achieves superior evaluation indexes in most situations. For spectral evaluation, the proposed algorithm has less spectral distortion and can better maintain the spectral resolution of HS images. In terms of spatial texture information, the high spatial resolution texture information of the PAN image is fully utilized to maintain the image textures. From the objective evaluation index, the proposed method can effectively reduce the spectral shift of HS images on the premise of maintaining a good spatial resolution.

3) Computation Complexity Analysis: Comparative results, in terms of computational complexity, are shown in Table V, where execution time estimates for each algorithm to run the pansharpening results are reported.

Although we can see that the proposed method is not that competitive in terms of computation efficiency. But the proposed method contains two steps, the first step is the training process of NSST-HMF, and the second step is the HS-pansharpening process. The computational complexity of the proposed method is mainly concentrated in the training

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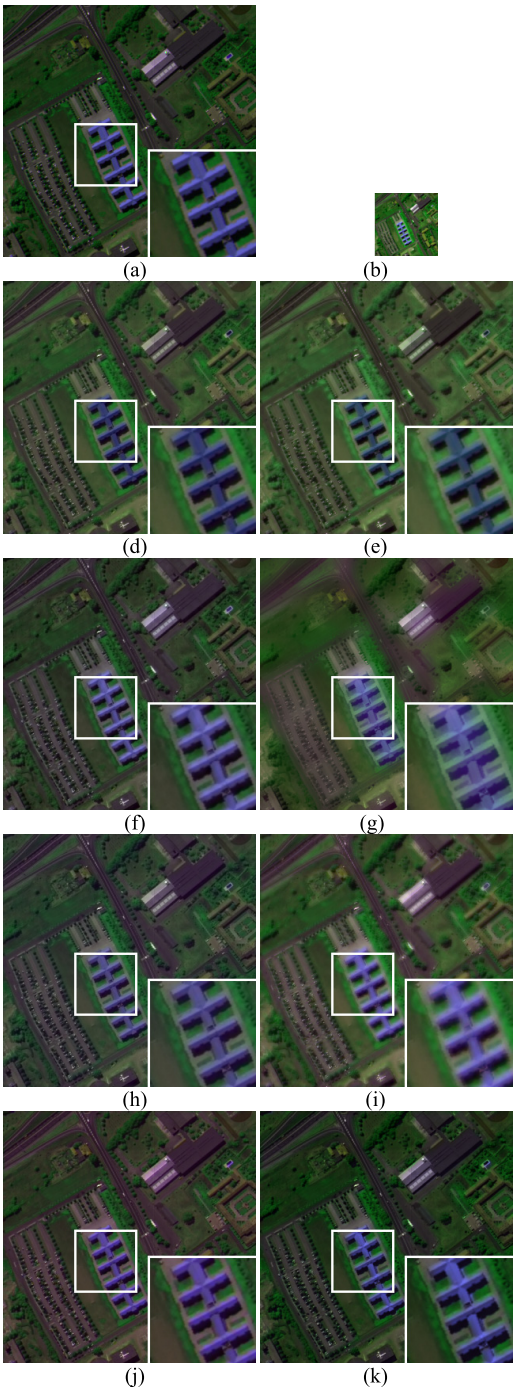


Fig. 14. Fusion results of the University of Pavia data set. (a) Reference image. (b) HS image. (c) GS. (d) PCA. (e) GSA. (f) GFPCA. (g) CNMF. (h) Wang *et al.* [44]. (i) Wang *et al.* [41]. (j) NSST-HMF.

process, the obtained HMF parameters can depict the multi-dimensional collaborative correlation of the HS image NSST coefficients, which can also be applied to other HS image applications such as HS image denoising and HS image change detection. Thus, we think although our method is time-consuming compared with classical HS-pansharpening methods, once the parameters have already been trained, our method can be applied to a broader application area with relatively short computation time.

TABLE II
QUALITY MEASURES FOR THE MOFFETT FIELD DATA SET

Methods	CC	SAM	RMSE	ERGAS
GS	0.9012	16.6479	19.6863	9.9605
PCA	0.8920	16.9522	20.1392	10.2427
GSA	0.9365	13.2269	13.1839	5.5834
GFPCA	0.9042	11.2143	47.2345	10.7512
CNMF	0.9453	9.9885	12.0433	5.5867
Ref.[45]	0.95114	8.7257	20.2738	6.4455
Ref.[42]	0.95555	12.7904	12.8937	5.7592
NSST-HMF	**0.96895**	**8.1018**	**11.3621**	**4.5669**

TABLE III
QUALITY MEASURES FOR THE CHIKUSEI DATA SET

Methods	CC	SAM	RMSE	ERGAS
GS	0.8772	8.4636	27.5356	6.4250
PCA	**0.9120**	10.8134	18.3417	6.4300
GSA	0.9021	12.9270	21.7896	6.8896
GFPCA	0.7923	14.6330	32.2957	8.3281
CNMF	0.8144	9.0409	14.7658	6.7966
Ref.[45]	0.7314	27.3595	53.7413	9.4791
Ref.[42]	0.8538	16.5232	30.1379	7.9133
NSST-HMF	0.8702	**8.2027**	**14.4887**	**6.1813**

TABLE IV
QUALITY MEASURES FOR THE UNIVERSITY OF PAVIA DATA SET

Methods	CC	SAM	RMSE	ERGAS
GS	0.9096	8.7217	21.2653	8.6523
PCA	0.8372	10.4424	29.2424	10.3504
GSA	0.9440	9.1903	14.5436	6.9874
GFPCA	0.8229	10.8755	26.6376	10.0870
CNMF	0.9272	7.9591	**9.9087**	**6.0804**
Ref.[45]	0.9243	8.9482	24.5860	9.4973
Ref.[42]	0.9305	15.8160	17.9048	8.2060
NSST-HMF	**0.9446**	**5.8656**	14.4014	6.2886

TABLE V
COMPARISON OF COMPUTATION TIME FOR DIFFERENT METHODS

Methods	Moffett field	Chikusei	University of Pavia
GS	1.51	2.91	1.13
PCA	1.71	5.17	1.77
GSA	1.72	3.19	1.26
GFPCA	1.31	1.47	1.14
CNMF	4.59	11.29	4.66
Ref. [43]	4318.12	17227.52	7929.97
Ref. [40]	35.62	145.53	65.89
NSST-HMF	219.82	511.42	297.96
Training	139.27	328.04	195.93
Pansharpening	80.55	183.38	102.03

C. Discussion

From the above experiments, it is possible to observe the following.

1) *To Compare With MGA HMT Model-Based Image Processing Methods:* the HMT model can depict the marginal statistical property of MGA subbands and the

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joint statistical property among them simultaneously, and the model parameters can be applied to a different image processing area. However, the MGA-HMT models can depict only the transfer relationship of images with only two dimensions through a tree structure. The proposed HMF model fully utilizes the multidirectional correlation of the HS image and improves the application accuracy, experimental results demonstrate this.

2) *To Compare With the HS-Pansharpening Methods:* in traditional HS-pansharpening methods, the strong correlation among the HS bands is not fully used which will lead to the strong spectral distortion phenomenon of the pansharpening results. The proposed HMF structure efficiently reduces the limitation.

VI. Conclusion

In this article, we first study the statistical properties of the HS image NSST coefficients in the spectral and spatial directions, and we obtain "clustering" and "aggregation" properties in the two directions. In addition, based on the above conclusions, the HMF model (NSST-HMF) of the HS image NSST coefficient is proposed to depict the multidimensional joint correlation of NSST coefficients. The correlations in the spatial and spectral directions are both considered by the prediction structure of HS hidden Markov to collaboratively predict the HMT coefficients, which provide a new and better way to extract and correlate the relationship of HS multiscale decomposition coefficients. Finally, a novel HS-pansharpening method is proposed based on the NSST-HMF model. The experimental results show that the proposed method can effectively improve the spatial resolution of the HS image while at the same time preserving the spectral features.

The proposed NSST-HMF model has several applications that can be investigated in the future. First, HS image denoising: Traditional MGA domain HMT models have shown great success in the area of image denoising. The noise component of the HS images usually affects the characteristics of the reflection spectrum and greatly reduces the accuracy of the data. Fully exploiting and utilizing the correlation among HS images will improve the denoising performance. Second, HS image change detection: The high spectral resolution of HS image provides the possibility for detecting subtle changes associated with land-cover transitions in the multitemporal images. However, the high-dimensional feature space and the high redundancy among channels make HS image change detection extremely challenging. The proposed NSST-HMF model can depict the transfer relationship along different directions of HS images.

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